INTERSECTION NUMBERS OF AMPLE INVERTIBLE SHEAVES ON A PROPER SCHEME ARE POSITIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites ten cross-chapter labels (including a definition and three section labels), each recorded in assembly.external_references with its canonical tag as a navigation pointer.

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Allowed standard prerequisites (editorial).

- Characterization of ample invertible sheaves by affine nonvanishing loci of sections. Stacks Project, Properties, Definition properties-definition-ample (tag 01PS). Produces the section s with X_s affine after replacing L_1 by a positive tensor power.
- Effective Cartier divisors and the intersection number reduction. Stacks Project, Varieties, Lemmas lemma-numerical-intersection-effective-Cartier-divisor (tag 0BEU) and lemma-intersection-number-in-terms-of-components (tag 0BES). Rewrite the intersection number on the divisor Z(s) and reduce to the integral case.
- Additivity of intersection numbers and finiteness of cohomology in dimension zero. Stacks Project, Varieties, Lemma lemma-intersection-number-additive (tag 0BER) and Lemma lemma-chi-tensor-finite (tag 0AYT). Allow replacing L_1 by a positive tensor power and supply the base case.

Source reference. Original lemma 0AYT: sources/stacks-chow/tags.txt, source lines 13801–13801. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0BES: sources/stacks-chow/tags.txt, source lines 14360–14360. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0BER: sources/stacks-chow/tags.txt, source lines 14359–14359. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01PS: sources/stacks-chow/tags.txt, source lines 2241–2241. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01WN: sources/stacks-chow/tags.txt, source lines 2488–2488. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Source reference. Original reference 01WQ: sources/stacks-chow/tags.txt, source lines 2491–2491. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Source reference. Original lemma 0A21: sources/stacks-chow/tags.txt, source lines 12689–12689. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0BEU: sources/stacks-chow/tags.txt, source lines 14362–14362. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X be proper over k. Let $Z \subset X$ be a closed subscheme of dimension d. If $\mathcal{L}_1, \ldots, \mathcal{L}_d$ are ample, then $(\mathcal{L}_1 \cdots \mathcal{L}_d \cdot Z)$ is positive.*

Proof. We will prove this by induction on d. The case $d = 0$ follows from Lemma 0AYT. Assume $d > 0$. By Lemma 0BES we may assume that Z is an integral closed subscheme. In fact, we may replace X by Z and $\mathcal{L}_i$ by $\mathcal{L}_i|_Z$ to reduce to the case $Z = X$ is a proper variety of dimension d. By Lemma 0BER we may replace $\mathcal{L}_1$ by a positive tensor power. Thus we may assume there exists a nonzero section $s \in \Gamma(X, \mathcal{L}_1)$ such that X_s is affine (here we use the definition of ample invertible sheaf, see Properties, Definition 01PS). Observe that X is not affine because proper and affine implies finite (Morphisms, Lemma 01WN) which contradicts $d > 0$. It follows that s has a nonempty vanishing scheme $Z(s) \subset X$. Since X is a variety, s is a regular section of $\mathcal{L}_1$, so $Z(s)$ is an effective Cartier divisor, thus $Z(s)$ has codimension 1 in X, and hence $Z(s)$ has dimension $d - 1$ (here we use material from Divisors, Sections 01WQ, 0C4S, and 0B3Q and from dimension theory as in Lemma 0A21). By Lemma 0BEU we have

$$(\mathcal{L}_1 \cdots \mathcal{L}_d \cdot X) = (\mathcal{L}_2 \cdots \mathcal{L}_d \cdot Z(s))$$

By induction the right hand side is positive and the proof is complete. □